

Uncovering the Shape of DNA by Diffraction Pattern

Doyun Lee,¹ Isaac Han,²

¹*Quarry Lane School*

²*B.S. in Physics from UCLA*

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ABSTRACT

The importance of DNA molecules as a matter of inheritance is widely taught in numerous high school and college classes. However, many students fail to understand the underlying principles and mechanisms of the helical structure of DNA. Despite the increased public awareness of female scientists, including Rosalind Franklin and her Photo 51, deeply involved Physical and Experimental concepts prevented students from developing the relationship between shape and x-ray imaging.

The importance of the DNA shape is demonstrated throughout the history of molecular biology: the paper published in 1953, *Molecular Structure of Nucleic Acids: Structure of Deoxyribose Nucleic Acid*, established a milestone in the scientific community, and its authors, James Watson and Francis Crick, and the collaborator, Maurice Wilkins, received the Nobel Prize [Wikipedia contributors \(2024a\)](#). Analyzing Photo 51 taken by Rosalind Franklin, the paper partially solves the central dogma of life, which explains the genetic information flow from DNA to produce protein [Wikipedia contributors \(2024b\)](#). Today, the mechanisms of DNA replication and Polymerase Chain Reaction (PCR) are well-studied by many, including high school students.

Historically, based on the X-ray diffraction pattern, the shape of the DNA was conjectured and verified [Wikipedia contributors \(2024b\)](#). The relationship between X-ray imaging technology and the DNA shape was discussed using the papers *Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid* and *X-ray Diffraction, Rosalind Franklin, and DNA*, by Watson and Crick and Rick Dower, respectively. This paper investigates the principle of DNA diffraction at the level of undergraduate student level.

Key words: DNA – Photo51 – X-ray Crystallography – Double Helix – Measurements

1 INTRODUCTION

1.1 DNA

DNA consists of a phosphate and sugar backbone with nitrogenous bases: adenine, thymine, cytosine, and guanine. Two strands of DNA chains form an antiparallel double-helix shape, and they are connected by hydrogen bonding between the nitrogenous bases. Nitrogenous bases could be categorized into purine (adenine and guanine) and pyrimidine (thymine and cytosine). For hydrogen bonding to occur, one purine and pyrimidine must be paired, revealed by the base-pairing rule where adenine and thymine are paired, and guanine and cytosine are paired [202 \(2022\)](#).

Van der Waals force plays a critical role in the stability of DNA molecules. Watson and Crick argue that the three-intertwined chain model with the phosphates near the fiber axis potentially leaving some bases outside is not plausible. The repulsive force near the phosphate is enough to overcome the van der Waals force in some regions due to the small Van der Waals distance, resulting in the destabilization of DNA molecules ([Watson & Crick 1953](#), p.1).

Oxygen plays an important role when it comes to Van der Waals force. Implied from the name “deoxyribose,” DNA lacks an oxygen atom in its sugar portion. Since oxygen has a high electronegativity, or the measure of strength attracting electrons, the presence of an oxygen molecule increases the van der Waals force, making the sugar more reactive and resulting in the potential breakdown of DNA. Removing

an oxygen atom decreases the van der Waals force, and DNA becomes more stable ([Barker et al. 2006](#), p.1).

1.2 X-ray Crystallography

Every material possesses distinct absorption rates, and their differences can be used to distinguish materials using X-ray, just as widely known in X-ray imagery or tumor histology. In X-ray crystallography, the crystalline units cause the incident X-ray beam to diffract in a specific direction. By measuring diffracted X-ray beams, certain characteristics of crystals, including the geometry, can be measured.

Although the detailed mechanism includes chemical compositions and wave mechanics analysis, the underlying principle can be simplified to Bragg’s law to determine the spacing between the crystalline structure, whose proof lies in simple geometry and wave mechanics.

Photos produced by X-ray diffraction were extensively analyzed by Watson and Crick to validate their DNA model. One pattern they suggested is a series of equally spaced along a helix shape from the photograph, today known as the double-helix structure [Dower \(2016\)](#).

1.3 Implications

The diagram, fig: 1, demonstrates the importance of simple distance metrics. The space between the center and the dark part on the top

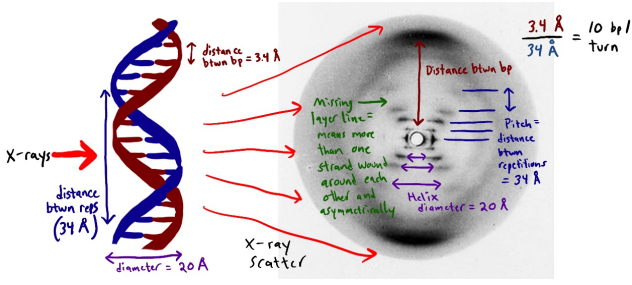


Figure 1. A simple diagram explaining how Photo 51 is related to the structure of DNA molecule. Taken from [Wikipedia](#)

indicates that the distance between repeating bases is 3.4 Årise/base-pair. If the distance between base pairs is 3.4 Åand the distance between pitches, the space between spots or the repeat of the helix, is 34 Å, one can calculate that there are 10 base pairs per rotation, by dividing 34 Åby 3.4 Å. By having 10 base pairs per rotation, one can also conclude the nature of rotation is 36 degrees per base pair since a full rotation is 360 degrees ?.

Having twisted double strands and base pairs pointing inside, the nature of inheritance can be inferred as well. Nucleotides only bond with their complementary bases – A to T and C to G. Together, they form information security - non-reactive deoxy-strands point the outside, and mutations can only occur when both pairs are replaced together by complementary pairs (Watson & Crick 1953, p.1).

2 EQUATIONS

For DNA modeling, it relies heavily on measurements. Equations associated with measurements are listed below

2.1 Bragg's Law

The diagram explaining how Photo 51, Fig. 1, captures the shape of a DNA molecule reveals the importance of measuring the distance between base pairs and pitches to capture the shape. If the distance between base pairs and the distance between pitches were 3.4Å and 34Å, respectively, one can calculate that there are 10 base pairs per rotation. Bragg's law is crucial for obtaining the measurements of DNA molecules ?.

One can infer local uniform geometry through Bragg's law. Experimenters can control the initial state of the light rays; the angle of the incident and the wavelength of the light used are given.

Assuming local uniformity, rays of parallel light would reflect upon hitting an atom. Since atoms in molecules are stable due to interactions with neighboring atoms, the angle of reflection is the same as the angle of incident, unlike the cases of Compton scatterings.

$$\theta_{\text{incident}} = \theta_{\text{reflection}} \quad (1)$$

where θ refers to the angle and subscripts specify which ones.

From Fig.2, With a little trigonometry, one can conclude that the ray reflected on the second layer would have traveled $2d \cdot \sin(\theta)$ more distance than the ray reflected on the first layer. Lights would experience constructive interference when they are in phase, which means that is the time when the extra distance traveled by the deeper layer in the crystal is multiples of their wavelength.

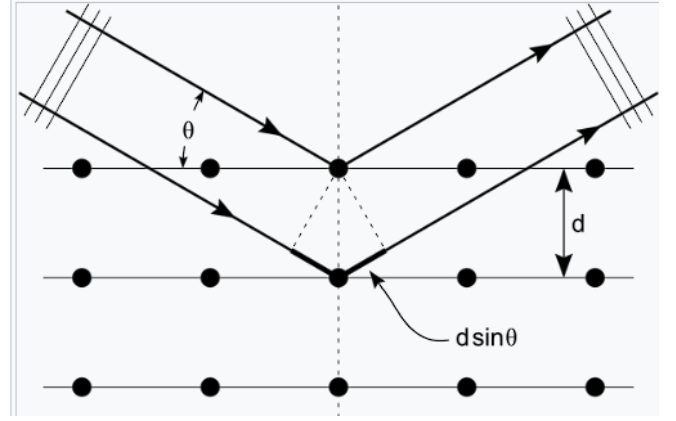


Figure 2. A simple diagram demonstrating Bragg's law. Taken from [Wikipedia](#)

$$n\lambda = 2d\sin(\theta) \quad (2)$$

where n is the diffraction order, d is the distance between the lattice, and θ is the angle of incident.

2.2 DNA absorption coefficient of various materials

From Appendix A, the nuclear bases typically pick up energy levels between 400 - 560 eV (electron Volt) whereas the DNA Phosphate and ATP group picks up energy levels between 2 - 82 eV ?. Although the specifics of experimental methods are different and the involved physics are different, it can be observed that different chemical behaves sensitively to the incident light's wavelength, from Appendix B. Especially in Appendix B, 2-Deoxy-D-Ribose has a significantly lower absorption spectrum compared to D-Ribose-S-Phosphate, which was the main discussion of the stability of DNA molecule in Watson and Crick's paper, where the presence of Oxygen molecule with high electronegativity can cause Van der Waals force.

In conclusion, different chemicals interact differently with the incident light based on its energy, which is famously known as the quantization of energy. It is shown in Appendix A and Appendix B that bases, ATP and Phosphate-Deoxy-Ribose backbones have different energy spectrums and the absorption cross-section falls rapidly. In short, with a correct X-ray scattering experiment, only the phosphate group whose electron density is high due to deoxidization would act as an opaque material, and all others can be considered transparent, due to their insignificant absorption cross-section.

2.3 Babinet's Principle

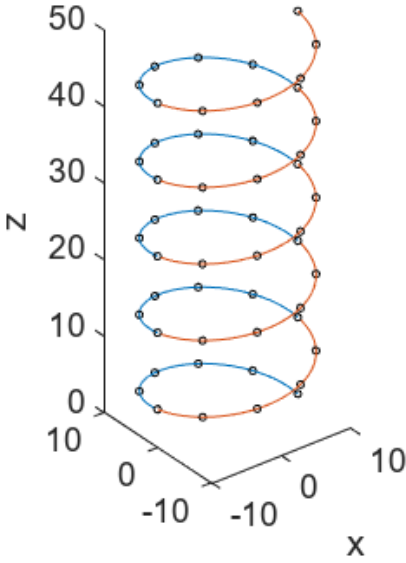
Babinet's principle states that the diffraction pattern from an opaque body is identical to a pigeonhole projection of the same size and shape, except for the beam intensity(Wikipedia). This principle simply states analyzing the diffraction pattern is the same as tracking the opaque body. In the case of DNA, the opaque body would be the Phosphate group.

3 MAIN - CHARACTERIZATION OF HELIX STRUCTURE

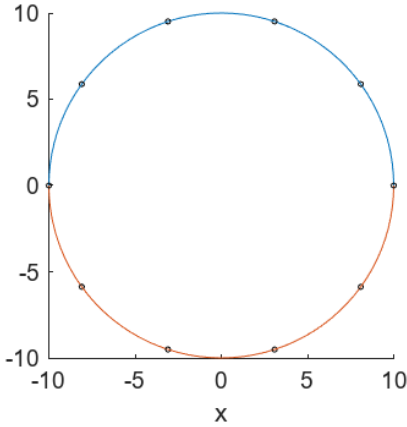
According to Dower (2016), The characterization of a single helix structure can be made with the following:

$$x = R \cos(2\pi t/T), y = R \sin(2\pi t/T), z = P(t/T)$$

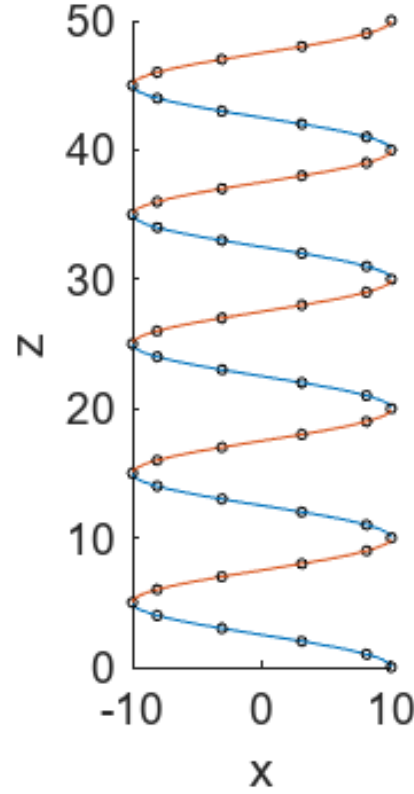
A demonstrative model is made with parameters $t = 0 : \pi/50 : 10\pi, T = 1, R = 10$.



Projection onto the x-y coordinate would result in the equation of the circle. As the parameter t increases, the function $P(t)$ determines how rapidly the helix deviates from the x-y plane.



According to Babinet's principle and discussion of the absorption spectrum of various materials, we can only consider the Phosphate backbone as the only opaque material during a well-coordinated X-ray spectroscopy experiment. One important characterization is that the projection of the helix structure can be thought of as the interference between two homogeneous grids, illustrated in the image below.



As demonstrated above, for a ray to be projected onto the screen, it has to pass through two sets of slanted lines whose magnitude of slopes is equal. A set of lines represents the frontal portion of a helix whereas the other set of lines represents the posterior portion of the line.

The tangent line slope of two sets of grids, marked by blue and red, can be mathematically derived using calculus. The result is taken from Dower (2016)'s tangential equation.1.

$$\begin{aligned} \text{Slope} &= \frac{dy}{dz} = R \left(\frac{2\pi}{P} \right) \cos\left(\frac{2\pi z}{P} \right) \\ &= R \left(\frac{2\pi}{P} \right) \end{aligned} \quad (3)$$

From Bragg's law and Dower (2016)'s eqn.2,

$$\sin(\theta_{\max}) = \frac{n\lambda}{d} \quad (4)$$

Measuring the interference pattern gives information about the maximum grating angle.

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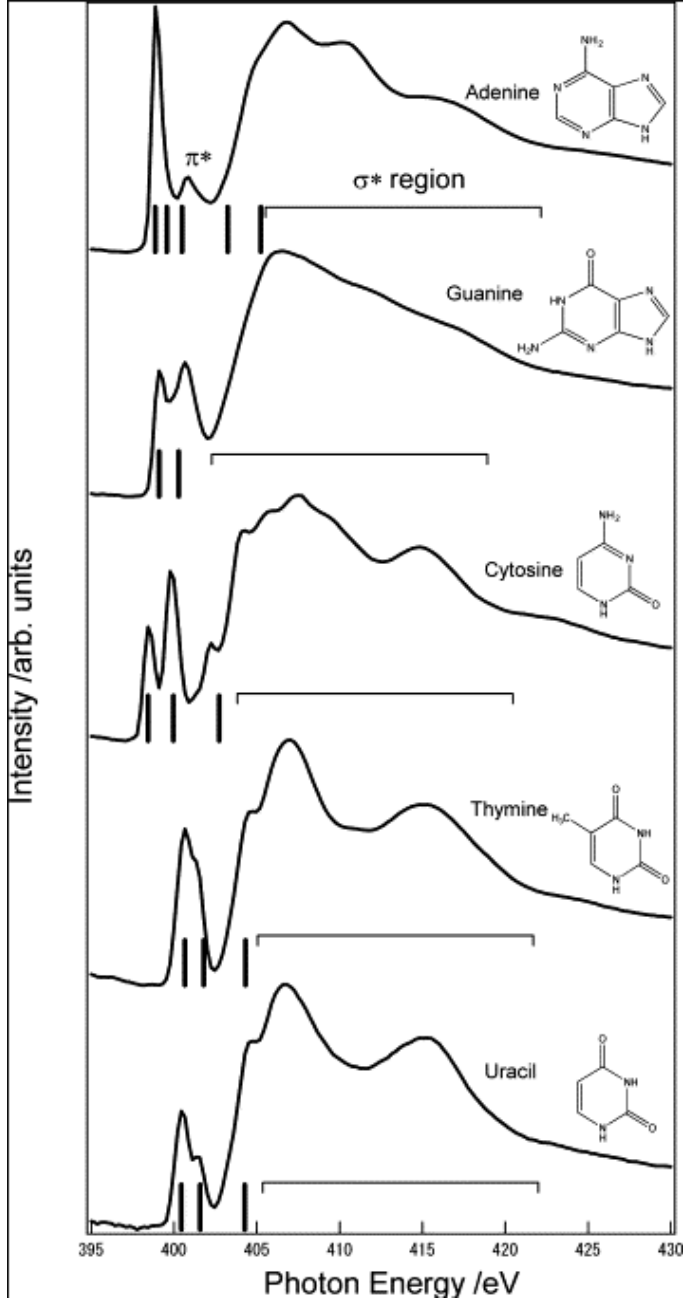
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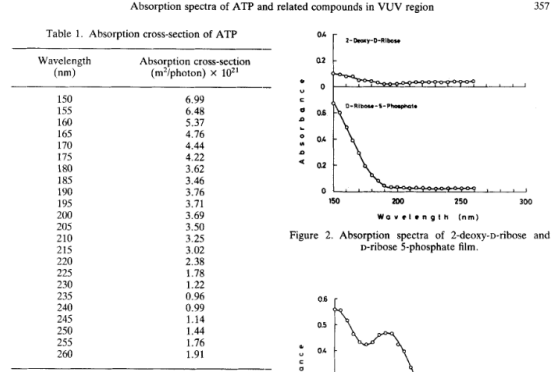
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APPENDIX A: ENERGY ABSORPTION GRAPH OF BASES TAKEN FROM FUJII AT EL



The image is taken from "X-ray absorption near edge structure of DNA bases around oxygen and nitrogen K-edge" by Fujii et al.

APPENDIX B: ENERGY ABSORPTION GRAPH OF ATP AND RIBOSE TAKEN FROM ATSUSHI ITO AND TAKASHI ITO



The image is taken from "ABSORPTION SPECTRA OF DEOXYRIBOSE, RIBOSEPHOSPHATE, ATP AND DNA BY DIRECT(150-190 nm) AND FAR-UV (190-260 nm) REGIONS USING SYNCHROTRON RADIATION AS A LIGHT SOURCE TRANSMISSION MEASUREMENTS IN THE VACUUM-UV" by Atsushi Ito and Takashi Ito.